Piron Romain | Ph.D

Postdoctoral researcher

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Tokyo, Japan | **Nationality:** French | **Languages:** English (Advanced), French (Native), 日本語 (Beginner)

Profile summary

- I am currently working on **physical modeling** and **certification** of quantum-networking platforms.
- Dual background in **theoretical physics and information theory**
- Based in Tokyo, with a growing interest in Japanese language and culture
- Detailed description of my scientific activities available on my website: https://rpiron.github.io/

Research awards

- **IEEE Quantum Week 2026 Best Paper Awards:** 2nd place in the quantum photonics track

Experience

Postdoctoral researcher | Channel estimation and numerical simulation

国立情報学研究所, Tokyo, Japan | September 2025 - Present

- **Physical modeling** of optical elements involved in photon-processing for quantum networks
- Development of **scalable simulation frameworks** of high-dimensional quantum optics models
- Algorithmic design for **quantum channel estimation** through **statistical inference**

Ph.D researcher | Quantum optimization and signal processing

CITI Laboratory, Lyon, France | October 2023 - October 2025

- Designed and implemented **quantum optimization algorithms** for **data inference in large-scale wireless networks**
- Validation through **Python simulations** and **experiments on quantum devices**
- **Interdisciplinary collaboration** at the intersection of quantum information and signal processing

Event coordinator | Volunteering in student's association

24h de l'INSA, Lyon, France | January 2024 - December 2024

- Led a team of 10 volunteers during **France's largest student-run festival**
- Developed strong **coordination, organization,** and **leadership skills** in a fast-paced environment

Research internship | Master's thesis in mathematical physics

CERN - European Organization for Nuclear Research, Geneva, Switzerland | February 2023 - July 2023

- Investigated **mathematical consistency conditions** in conformal field theory using the **bootstrap method**, with implications for the AdS/CFT correspondence

Education

Ph.D in Signal processing and optimization

INSA Lyon (Claude Bernard Lyon 1 University), Lyon, France | October 2025

Thesis: Quantum Algorithms for Non-Orthogonal Multiple Access Systems

M.Sc in Physics - Double degree

EPFL (Ecole Polytechnique Fédérale de Lausanne), Lausanne, Switzerland | October 2023

Advanced coursework in quantum field theory, general relativity, and high-energy physics.

M.Sc in Engineering

CentraleSupélec (Paris Saclay University), Gif-sur-Yvette, France | October 2023

Mathematics and physics with interdisciplinary exposure from financial modeling to quantum technologies

Selected publications

- R.Piron, A.Soeda, "Parameter Calibration for Reduced-Bandwidth Two-Photon Waveguide-QED Simulations", to appear in in the Proceedings of the **2026 IEEE International Conference on Quantum Computing and Engineering**
- R.Piron, C.Goursaud, "*Quantum Annealing for Active User Detection in NOMA Systems*", Proceedings of the **2024 58th Asilomar Conference on Signals, Systems, and Computers**